1. Create a directory named lab6 and a text file named permissions.txt inside it.

```
akansha@localhost:~$ mkdir lab6
akansha@localhost:~$ touch lab6/permissions.txt
akansha@localhost:~$ ls -l lab6/permissions.txt
-rw-r--r--. 1 akansha akansha 0 Oct 31 11:30 lab6/permissions.txt
```

2. Display the long listing of permissions.txt and identify the owner, group and current rwx permissions.

```
akansha@localhost:~$ ls -l lab6/permissions.txt
-rw-r--r--. 1 akansha akansha 0 Oct 31 11:30 lab6/permissions.txt

Identification:
- File Type & Permissions: -rw-r--r-- (File, Owner: rw-, Group: r--, Others:
r--)
- Owner: akansha
- Group: akansha
```

3. Change the permissions of permissions.txt using symbolic notation so that the owner has read, write and execute permissions, the group has read and write permissions, and others have no permissions.

```
akansha@localhost:~$ chmod u=rwx,g=rw,o= lab6/permissions.txt
akansha@localhost:~$ ls -l lab6/permissions.txt
-rwxrw----. 1 akansha akansha 0 Oct 31 11:30 lab6/permissions.txt
```

4. Change the permissions of permissions.txt using numeric mode 640 and verify the result.

```
akansha@localhost:~$ chmod 640 lab6/permissions.txt
akansha@localhost:~$ ls -l lab6/permissions.txt
-rw-r-----. 1 akansha akansha 0 Oct 31 11:30 lab6/permissions.txt
```

5. Change the permissions of permissions.txt using numeric mode 755 and verify the result.

```
akansha@localhost:~$ chmod 755 lab6/permissions.txt
akansha@localhost:~$ ls -l lab6/permissions.txt
-rwxr-xr-x. 1 akansha akansha 0 Oct 31 11:30 lab6/permissions.txt
```

6. Remove execute permission from the owner of permissions.txt using symbolic chmod notation.

```
akansha@localhost:~$ chmod u-x lab6/permissions.txt
akansha@localhost:~$ ls -l lab6/permissions.txt
-rw-r-xr-x. 1 akansha akansha 0 Oct 31 11:30 lab6/permissions.txt
```

7. Add read and execute permissions for the group of permissions.txt without changing the owner's permissions.

```
akansha@localhost:~$ chmod g+rx lab6/permissions.txt
akansha@localhost:~$ ls -l lab6/permissions.txt
-rw-r-xr-x. 1 akansha akansha 0 Oct 31 11:30 lab6/permissions.txt
```

8. Remove all permissions from others for permissions.txt and verify the result.

```
akansha@localhost:~$ chmod o-rwx lab6/permissions.txt
akansha@localhost:~$ ls -l lab6/permissions.txt
-rw-r-x---. 1 akansha akansha 0 Oct 31 11:30 lab6/permissions.txt
```

9. Create a directory named securedata and set its permissions to 750 using numeric notation.

```
akansha@localhost:~$ mkdir securedata
akansha@localhost:~$ chmod 750 securedata
akansha@localhost:~$ ls -ld securedata
drwxr-x---. 2 akansha akansha 6 Oct 31 11:32 securedata
```

10. Explain and demonstrate the difference between file and directory read, write and execute permissions by testing access to a practice directory.

```
Explanation:
- File Permissions:
  r (Read): Read file contents.
  w (Write): Modify file contents.
  x (Execute): Execute file as a script/program.

- Directory Permissions:
  r (Read): List files inside directory (e.g., ls).
  w (Write): Create, delete, or rename files inside directory.
  x (Execute): Enter directory (cd) and access metadata/files inside.

Demonstration:
akansha@localhost:~$ mkdir testdir && touch testdir/file.txt
akansha@localhost:~$ chmod 400 testdir # Read-only, no execute (cannot cd into
it)
akansha@localhost:~$ ls testdir
file.txt
akansha@localhost:~$ cd testdir
bash: cd: testdir: Permission denied
```

11. Display the current umask value and record the result.

```
akansha@localhost:~$ umask
0022
```

12. Temporarily change the umask value to 027 and create a new file and directory. Observe and verify their resulting permissions.

```
akansha@localhost:~$ umask 027
akansha@localhost:~$ touch umask_file.txt
akansha@localhost:~$ mkdir umask_dir
akansha@localhost:~$ ls -l umask_file.txt
-rw-r-----. 1 akansha akansha 0 Oct 31 11:35 umask_file.txt
akansha@localhost:~$ ls -ld umask_dir
drwxr-x---. 2 akansha akansha 6 Oct 31 11:35 umask_dir

Analysis:
Base File: 666 - 027 = 640 (-rw-r-----)
Base Directory: 777 - 027 = 750 (drwxr-x---)
```

13. Restore the umask to its previous value and verify the current umask.

```
akansha@localhost:~$ umask 0022
akansha@localhost:~$ umask
0022
```

14. Create a new user and group for ACL practice, then create a file named aclfile.txt for testing.

```
akansha@localhost:~$ sudo useradd acluser
akansha@localhost:~$ sudo groupadd aclgroup
akansha@localhost:~$ touch aclfile.txt
akansha@localhost:~$ ls -l aclfile.txt
-rw-r--r--. 1 akansha akansha 0 Oct 31 11:38 aclfile.txt
```

15. Display the current ACL of aclfile.txt using getfacl and identify the owner, group, user, group and mask entries.

```
akansha@localhost:~$ getfacl aclfile.txt
# file: aclfile.txt
# owner: akansha
# group: akansha
user::rw-
group::r--
other::r--
```

16. Use setfacl to give a specified user read and write permission on aclfile.txt without changing the normal owner/group permission bits. Verify using getfacl.

```
akansha@localhost:~$ setfacl -m u:acluser:rw aclfile.txt
akansha@localhost:~$ getfacl aclfile.txt
# file: aclfile.txt
# owner: akansha
# group: akansha
user::rw-
user:acluser:rw-
group::r--
mask::rw-
other::r--

akansha@localhost:~$ ls -l aclfile.txt
-rw-rw-r--+ 1 akansha akansha 0 Oct 31 11:38 aclfile.txt
```

17. Use setfacl to give a specified group read permission on aclfile.txt and verify the ACL.

```
akansha@localhost:~$ setfacl -m g:aclgroup:r aclfile.txt
akansha@localhost:~$ getfacl aclfile.txt
# file: aclfile.txt
# owner: akansha
# group: akansha
user::rw-
user:acluser:rw-
group::r--
group:aclgroup:r--
mask::rw-
other::r--
```

18. Modify and remove a specific ACL entry from aclfile.txt using setfacl. Verify the changes using getfacl.

```
# Modify acluser ACL to read-only:
akansha@localhost:~$ setfacl -m u:acluser:r aclfile.txt

# Remove aclgroup ACL entry:
akansha@localhost:~$ setfacl -x g:aclgroup aclfile.txt

akansha@localhost:~$ getfacl aclfile.txt
# file: aclfile.txt
# owner: akansha
# group: akansha
user::rw-
user:acluser:r--
group::r--
mask::r--
other::r--
```

19. Set a default ACL on a directory so that newly created files inside it inherit the required access permissions. Create a test file and verify the inherited ACL.

```
akansha@localhost:~$ mkdir acldir
akansha@localhost:~$ setfacl -m d:u:acluser:rw acldir
akansha@localhost:~$ getfacl acldir
# file: acldir
# owner: akansha
# group: akansha
user::rwx
group::r-x
other::r-x
default:user::rwx
default:user:acluser:rw-
default:group::r-x
default:mask::rw-
default:other::r-x

akansha@localhost:~$ touch acldir/inherited_file.txt
akansha@localhost:~$ getfacl acldir/inherited_file.txt
# file: acldir/inherited_file.txt
# owner: akansha
# group: akansha
user::rw-
user:acluser:rw-
group::r-x
mask::rw-
other::r--
```

20. Create, modify, verify and clean up file permissions and ACLs in a complete workflow using chmod, umask, setfacl and getfacl.

```
akansha@localhost:~$ mkdir workflow_dir
akansha@localhost:~$ touch workflow_dir/file.txt
akansha@localhost:~$ chmod 750 workflow_dir
akansha@localhost:~$ chmod 640 workflow_dir/file.txt
akansha@localhost:~$ setfacl -m u:acluser:rw workflow_dir/file.txt
akansha@localhost:~$ getfacl workflow_dir/file.txt
# Cleanup step:
akansha@localhost:~$ setfacl -b workflow_dir/file.txt
akansha@localhost:~$ rm -rf workflow_dir acldir securedata lab6 aclfile.txt
umask_file.txt umask_dir testdir
akansha@localhost:~$ sudo userdel -r acluser 2>/dev/null
akansha@localhost:~$ sudo groupdel aclgroup 2>/dev/null
```